google, ibm, ms

Projects suggestions (business aligned projects)

1. Demand forecasting and time series modelling

Approach

(1) Pick real-dataset

e.g sales data, energy consumption, flight, website traffic, weather data

(2) Start with exploration

- look for trends, seasonality (weekly, monthly, yearly patterns)
- look for anomalies and understand why they occured

(3) build multiple models

- use classical method e.g ARIMA or SARIMA
- use a modern baseline e.g Prophet, Meta forecasting library which handles seasonality
- deep learning approach e.g LSTM

be able to explain why one model outperforms the other?

articulate trade-offs between interpretability and accuracy

identify where each model fails

connect forecast to business impact

explain how better predictions improve costs, staffing decisions or prevent outages

Quick Tip:

> What to do if there were a real fraction system

> How often would you retrain it

> How will you trigger an alert